Sin/Cos Positional Encoder

Vector of Dimension "2" to define Position

$$\begin{bmatrix} \sin(\omega t) \\ \cos(\omega t) \end{bmatrix}$$

Where "t" is the step (position) and ω is the frequency of trigonometric function Sine/Cosine

This vector can be used to define word position at step "t" , t=0,1,2,3,4,5,6,7,....

- **Values are between -1 and 1** ***(bounded values)***

  Distance between vectors representing positions "t" and "t+1" is constant irrespective of "t" [Distance (Pos1, Pos2) = Distance (Pos90, Pos91)]

- ***"t" may take any positive value*** ***(No limitation of length of Sequence)***

- **Position Vector Length = 2 (Cannot be combined with Word Embedding Vector) ??**